

Original-divisor descent from rational Erdős–Straus trace sources

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21 September 2026

Abstract

We determine every same-word return obtained by replacing an original residual by a divisor, starting from rational positive ES tails with integral first trace. The word budget survives exactly at a specified residue class modulo $4 \prod_l l^{\lceil v_l(u)/2 \rceil}$. A one-step integral return is guaranteed for every such source precisely for the nine words dividing 36. Outside this list we construct infinitely many hard-prime sources on which the entire residual-divisor system, including the middle complement, fails. All source multiplicities and denominator inverses are retained. This is a source-descent and obstruction theorem, not a proof that every prime supplies an initial ES state or a suitable rational trace source.

Exact same-word residual-divisor descent

Every map, congruence, inverse parameter and lost coordinate is retained.

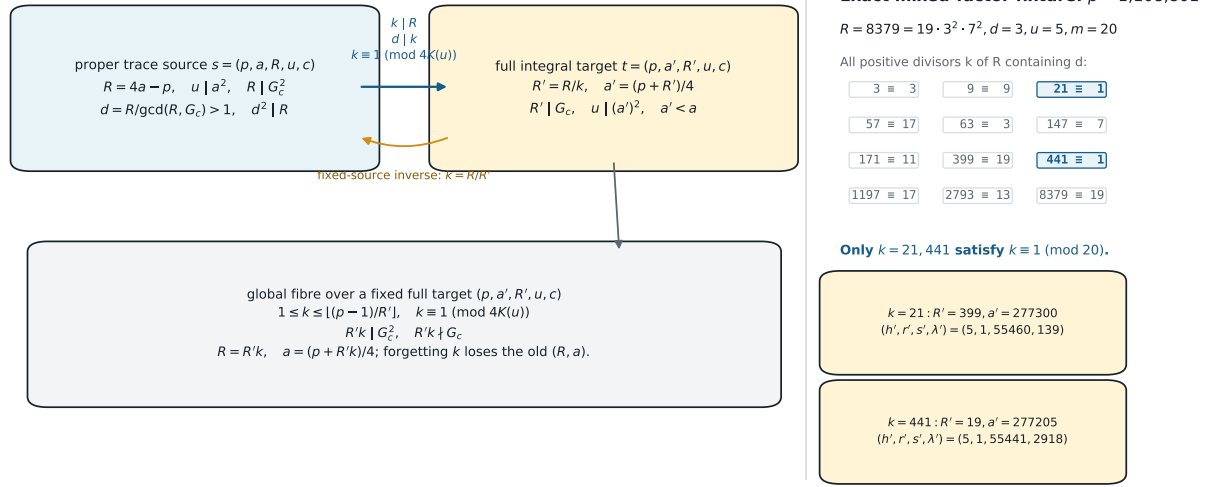


Figure 1: The exact source–target map of Theorem 3.1, its fixed-source inverse, the full global fibre, and the mixed-factor fixture of equation (18). The script recomputes every displayed integer before rendering.

1 Scope: a composite-source descent, not initial occupancy

The preceding prime-shell trace theorem [3] supplies an integer return from an integral rational first trace when the distinguished denominator is prime. It explicitly leaves the composite case open. Here the distinguished denominator may be arbitrary. We classify every residual-divisor return that keeps its original divisor word. Nine words have a uniform one-step return; for every other fixed word, infinitely many hard primes give exact obstructions within this transformation system. The original prime is never changed.

The exact nine-word boundary—and what fails outside it

A uniform one-step square deletion exists iff $H_{4K(u)} = \{1\}$ iff $u \mid 36$.

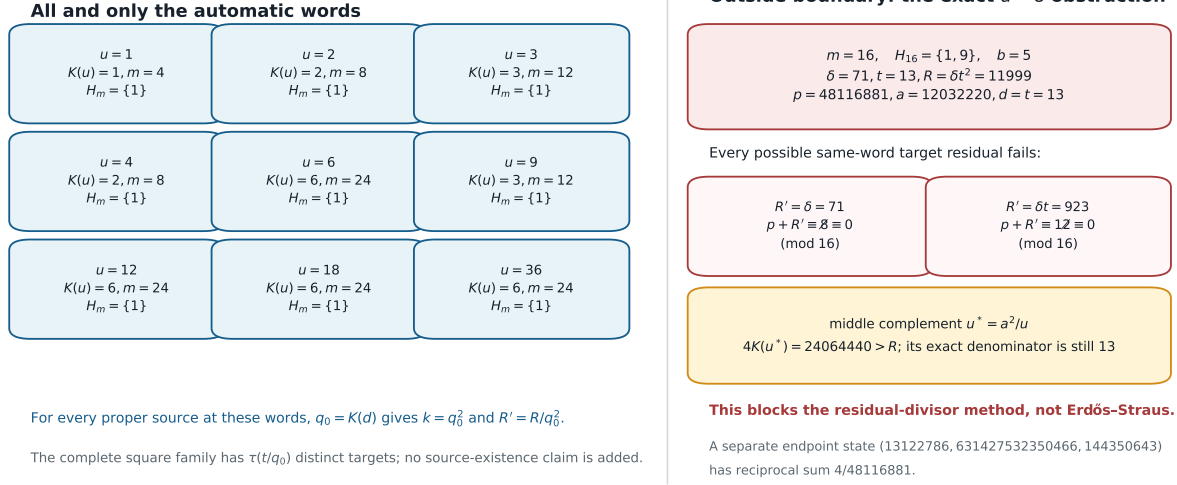


Figure 2: The square-image mechanism behind Theorems 4.2 and 7.2. The right panel is the complete $u = 8$ terminal fixture; it obstructs this descent method while retaining an explicit ES endpoint.

This remains the programme’s Turn 7. No theorem below asserts that a prescribed prime supplies the initial rational trace candidate. The Type-I/II formulas are classical [1, §2, Propositions 2.2 and 2.6]; the trace relaxation is inherited from [4, TR17–19]. The finite capacity bound used in the optimality theorem is from [5, Theorem “Sharp finite-repair bound”] and is reproved here. No historical-priority determination is made. Here a *hard prime* means

$$p \bmod 840 \in \{1, 121, 169, 289, 361, 529\}.$$

The counterfamily below lies in the first of these classes.

2 The full original divisor box and its rational trace relaxation

Fix a prime $p \equiv 1 \pmod{4}$, an integer $p/4 < a < p/2$, and $R = 4a - p$. Thus $3 \leq R < p$, $R \equiv 3 \pmod{4}$ and $\gcd(pa, R) = 1$. Retain the integer word and every exponent:

$$a = \prod_l l^{e_l}, \quad u = \prod_l l^{f_l} \mid a^2, \quad 0 \leq f_l \leq 2e_l, \quad \beta_l = f_l - e_l.$$

For $c \in \{E, M\}$ define

$$G_E = 4u + 1, \quad G_M = p + 4u.$$

Before requiring the full gate $R \mid G_c$, these original words give the positive rational denominator triples

$$E : \left(a, \frac{pa + a^2/u}{R}, \frac{pa + p^2u}{R} \right), \tag{1}$$

$$M : \left(a, \frac{p(a + a^2/u)}{R}, \frac{p(a + u)}{R} \right). \tag{2}$$

They satisfy $4/p = 1/x + 1/y + 1/z$ by multiplication.

Lemma 2.1 (Exact trace domain). *The tail trace $y + z$ is integral exactly when $R \mid G_c^2$. The two reduced tail denominators then equal*

$$d = \frac{R}{\gcd(R, G_c)}, \quad d^2 \mid R. \quad (3)$$

They are integers exactly when $d = 1$.

Proof. The traces are respectively $(a + pu)^2/(Ru)$ and $p(a + u)^2/(Ru)$. Since $u \mid a^2$, both $u \mid (a + pu)^2$ and $u \mid (a + u)^2$. All p, a, u are units modulo R , in particular $\gcd(u, R) = 1$. Since $p \equiv 4a \pmod{R}$, $a + pu \equiv a(1 + 4u)$ and $4(a + u) \equiv p + 4u$ there. The exact tail denominators in (1) and (2) follow by multiplying their integer numerators by the unit u when needed. If $l^e \parallel R$ and $g = v_l(G_c)$, the exponent of l in the reduced denominator is $\max(e - g, 0)$. The trace gate $R \mid G_c^2$ gives $e \leq 2g$, hence $2\max(e - g, 0) \leq e$. This proves $d^2 \mid R$ prime by prime. $\square$

A record with $d > 1$ will be called a proper trace source. It contains an *available integer divisor word*, but the full residual gate is not yet satisfied. It is different from the earlier raw integer sum-gate source in [4, TR7–16], which imposed its full gate before testing a trace.

These sources also recover every rational integral-trace tail pair. Let $S = y + z \in \mathbb{Z}$ and put $b = Ry - pa$, $c = Rz - pa$. The reciprocal identity gives

$$yz = \frac{paS}{R}, \quad b + c = RS - 2pa \in \mathbb{Z}, \quad bc = (pa)^2 \in \mathbb{Z}.$$

Thus the rational numbers b, c are roots of the monic polynomial

$$X^2 - (RS - 2pa)X + (pa)^2 \in \mathbb{Z}[X].$$

A rational algebraic integer is an integer. Moreover $1/y, 1/z < R/(pa)$, so $b, c > 0$. The three possible pairs of p -valuations are $(0, 2), (1, 1), (2, 0)$. For completeness, write the common reduced denominator as $y = A/d, z = C/d$; it is common because $z = S - y$. Then $RAC = paSd^2$ and $\gcd(AC, d) = 1$, so $d^2 \mid R$. If $R = d^2r$, then $rAC = paS$; since $\gcd(r, pa) = 1$, write $S = rT$. The equality $AC = paT$ and the facts $\gcd(AC, d) = \gcd(pa, d) = 1$ give $\gcd(T, d) = 1$. Consequently $\gcd(R, S) = r$ and

$$d^2 = R/\gcd(R, S).$$

After retaining an E orientation, the first pair has $b = a^2/u, c = p^2u$; the middle pair has $b = pa^2/u, c = pu$. Their inverses are exactly (1)–(2). Both middle tail orientations remain separate.

For any positive integer n write

$$K(n) = \prod_{l^e \parallel n} l^{\lceil e/2 \rceil}, \quad B(n) = \prod_{l^e \parallel n} l^{\lfloor e/2 \rfloor}.$$

The actual word budget is $u \mid a^2 \iff K(u) \mid a$. Put $m = 4K(u)$. Then

$$p + R \equiv 0 \pmod{m}, \quad \gcd(pR, m) = 1. \quad (4)$$

No absent prime coordinates are adjoined to this budget.

3 Every same-word residual-divisor return

Theorem 3.1 (Complete divisor-removal domain). *For a trace source (p, a, R, u, c) , the complete set of original same-channel returns that keep p, u and replace R by a divisor is indexed by the positive integers*

$$\mathcal{K} = \{k \in \mathbb{Z}_{>0} : k \mid R, d \mid k, k \equiv 1 \pmod{m}\}. \quad (5)$$

The map and the inverse after this source is retained are

$$R' = R/k, \quad a' = (p + R')/4, \quad u' = u; \quad k = R/R'. \quad (6)$$

For a proper source every return has $a' < a$ and hence terminates in one strictly decreasing step at an original integral state.

Proof. The full new gate $R/k \mid G_c$ is equivalent prime by prime to $d \mid k$. The new integral distinguished denominator and its word budget together are $p + R/k \equiv 0 \pmod{m}$. Multiplying by k and using (4) proves that this is equivalent to $k \equiv 1 \pmod{m}$. This condition includes $k \equiv 1 \pmod{4}$, so $R' \equiv 3 \pmod{4}$ and $R' \geq 3$. Also $R' \leq R < p$; hence $p/4 < a' < p/2$. The condition $K(u) \mid a'$ proves $u \mid a'^2$ in the actual integers. If $d > 1$ then $k \geq d > 1$ and $a' < a$. All steps are equivalences, proving completeness. $\square$

The full channel-labelled output is computed, not inherited unchanged from the rational tails. The label order records which tail recovers u ; it is not a claim that the denominators increase numerically. Put

$$g' = \gcd(a', u), \quad h' = g'^2/u, \quad r' = u/g', \quad s' = a'/g'.$$

Prime valuations prove $a' = h'r's'$, $u = h'r'^2$ and $\gcd(r', s') = 1$. For E, set $\kappa' = (pr' + s')/R'$ and return $(a', h's'\kappa', ph'r'\kappa')$. For M, set $\lambda' = (r' + s')/R'$ and return $(a', ph's'\lambda', ph'r'\lambda')$. The full gates prove the quotient integrality. Their labelled divisor inverses are $a'^2/(R'y - pa')$ and $pa'^2/(R'y - pa')$ respectively. The centred prime-exponent vectors satisfy

$$\beta'_l = \beta_l + v_l(a) - v_l(a').$$

A numerical word may survive even though the distinguished denominator's other prime factors change. Those old and new prime coordinates remain recorded.

3.1 Complete inverse fibres and integer coefficient maps

Fix a full original target (p, R', a', u, c) satisfying $R' = 4a' - p$, $3 \leq R' < p$, $R' \equiv 3 \pmod{4}$, $u \mid (a')^2$, and $R' \mid G_c$. Every proper input mapping to it is recovered by the positive integers

$$1 \leq k \leq \left\lfloor \frac{p-1}{R'} \right\rfloor, \quad k \equiv 1 \pmod{m}, \quad R'k \mid G_c^2, \quad R'k \nmid G_c, \quad (7)$$

with $R = R'k$ and $a = (p + R)/4$. The old word budget follows from $k \equiv 1 \pmod{m}$. Indeed $p + R' \equiv 0 \pmod{m}$ at the target, while $p + R'k = (p + R') + R'(k - 1) \equiv 0 \pmod{m}$. Thus a is an integer and $u \mid a^2$. The bound and $k \equiv 1 \pmod{4}$ give $0 < R < p$ and $R \equiv 3 \pmod{4}$; the last two divisibilities in (7) give the trace gate and properness. Finally $R' \mid G_c$ implies, prime by prime, that the deletion denominator $d = R/\gcd(R, G_c)$ divides k . These conditions are therefore both necessary and sufficient, including all original exponent multiplicities.

For exact fibre typing, let

$$\mathcal{A} = \{(s, k) : s \text{ is a proper source and } k \in \mathcal{K}_s\}, \quad \mathcal{T}_{\text{hit}} = F(\mathcal{A}),$$

where $F(s, k)$ is the target in (6). One source can have several arrows and targets; for example $(p, a, R, u, c) = (101, 41, 63, 1, M)$ has $k = 9$ and $k = 21$. The correctly typed pushforward is

$$F_* : \mathbb{Z}^{(\mathcal{A})} \longrightarrow \mathbb{Z}^{(\mathcal{T}_{\text{hit}})}, \quad [\alpha] \longmapsto [F(\alpha)].$$

Choose one arrow $\alpha_t \in F^{-1}(t)$ for every hit target. Then

$$\ker F_* = \bigoplus_{t \in \mathcal{T}_{\text{hit}}} \left\langle [\alpha] - [\alpha_t] : \alpha \in F^{-1}(t) \setminus \{\alpha_t\} \right\rangle_{\mathbb{Z}}.$$

The chosen arrows define a section $s([t]) = [\alpha_t]$. The actual integral isomorphism retaining every arrow is

$$x \mapsto (F_*x, x - sF_*x), \quad \mathbb{Z}^{(\mathcal{A})} \cong \mathbb{Z}^{(\mathcal{T}_{\text{hit}})} \oplus \ker F_*,$$

with inverse $(y, z) \mapsto s(y) + z$. After the source and fibre-difference coordinates are discarded, no inverse remains.

There is a finite coefficient calculation before any successful word is selected. Use the group ring of $G_m = (\mathbb{Z}/m\mathbb{Z})^\times$, with $[g]$ for its basis elements. The integer d and every prime dividing R/d are units modulo m , because $\gcd(R, m) = 1$. Since $k = dv$ in (5), the labelled pushforward is

$$\pi_{R,d} : \mathbb{Z}^{(\text{Div}(R/d))} \longrightarrow \mathbb{Z}[G_m], \quad e_v \mapsto [v \bmod m].$$

The group-ring image alone does not retain the divisor label v ; the labelled certificate does. The complete integer count is

$$|\mathcal{K}| = \text{coeff}_{[d^{-1}]} \prod_{l^e \parallel R/d} ([1] + [l] + \cdots + [l^e]). \quad (8)$$

Each factor is geometric: its $e + 1$ exponent choices occur once each. It is not an expansion of $([1] + [l])^e$ which would introduce coloured multiplicities. The integer coefficient and the original divisors representing it are retained.

If a target word exists, one exists with

$$\Omega(k/d) \leq |G_m| - 1 = \varphi(m) - 1. \quad (9)$$

Choose a shortest original prime-occurrence word for d^{-1} . Its successive partial products, starting with one, are distinct; a repeated partial product would let an identity subword be removed without exceeding any exponent bound. This proves (9). It bounds an existing word; it does not assert that the target coefficient is nonzero.

4 Square removal and the exact nine-seed guarantee

Write

$$R = \delta t^2, \quad \delta = \prod_{l^e \parallel R} l^{e \bmod 2}, \quad q_0 = K(d), \quad T = t/q_0.$$

The squarefree part δ is not the radical, and it need not be coprime to t . The fact $d^2 \mid R$ proves $q_0 \mid t$.

Proposition 4.1 (Every square deletion and its capacity). *For $q \in \mathbb{Z}_{>0}$, the square members $k = q^2$ of (5) are precisely*

$$q \mid t, \quad q_0 \mid q, \quad q^2 \equiv 1 \pmod{m}. \quad (10)$$

Let $H_m = \{x^2 : x \in G_m\}$. Their complete count is

$$\text{coeff}_{[q_0^{-2}]} \prod_{l^e \parallel T} ([1] + [l^2] + \cdots + [l^{2e}]) \quad \text{in } \mathbb{Z}[H_m]. \quad (11)$$

If nonzero, it has a word with $\Omega(q/q_0) \leq |H_m| - 1$.

Proof. The two divisibilities are exactly $q^2 \mid R$ and $d \mid q^2$. Write $q = q_0 v$, with $v \mid T$, for the coefficient formula. The preceding partial-product proof applies to the squared residue of each original prime. Equivalently the labelled map is $e_v \mapsto [v^2 \bmod m]$ from $\mathbb{Z}^{(\text{Div}(T))}$ to $\mathbb{Z}[H_m]$. $\square$

For

$$m = 2^\alpha \prod_{i=1}^r l_i^{e_i}, \quad \alpha \geq 2,$$

where the l_i are distinct odd primes,

$$|H_m| = 2^{\max(0, \alpha-3)} \prod_{i=1}^r \frac{l_i^{e_i-1}(l_i-1)}{2}. \quad (12)$$

At an odd prime power, $(x-1)(x+1) \equiv 0 \pmod{l_i^{e_i}}$ has just the two unit solutions ± 1 . At modulus four there are two; at 2^α , $\alpha \geq 3$, there are four, since one of $v_2(x-1), v_2(x+1)$ is one and the other must be at least $\alpha-1$. CRT and the square-map kernel give (12). Consequently

$$H_m = \{1\} \iff m \mid 24 \iff K(u) \mid 6 \iff \boxed{u \mid 36}. \quad (13)$$

Theorem 4.2 (Uniform one-step descent for nine exact words). *For every prime $p \equiv 1 \pmod{4}$, every proper E or M trace source with*

$$u \in \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$$

has an original same-channel integral return at

$$R' = R/K(d)^2, \quad a' = (p + R')/4, \quad u' = u, \quad a' < a. \quad (14)$$

Its complete square-deletion family has exactly

$$\tau(T) = \prod_{l^e \parallel R} \left(\left\lfloor \frac{e}{2} \right\rfloor - \left\lceil \frac{v_l(d)}{2} \right\rceil + 1 \right) \quad (15)$$

distinct original targets.

Proof. All the actual square factors are units modulo m . Equation (13) therefore makes every choice in (10) available, including $q = q_0$. Here $\tau(T)$ is the number of positive divisors of T , with $v_l(T) = \lfloor v_l(R)/2 \rfloor - \lceil v_l(d)/2 \rceil$. The target count is the number of divisors of T . Distinct q give distinct R' and hence distinct distinguished denominators. Theorem 3.1 proves integrality, positivity and strict descent. $\square$

This theorem does not assert the existence of a proper trace source at every prime. For hard primes its E part is vacuous for these nine words: the factors 2 and 3 are residues, whereas a relaxed E gate has negative word character. The M part is nonvacuous. The theorem's full domain is all $p \equiv 1 \pmod{4}$.

5 The complement is an available word only on its exact gate domain

At a fixed original a , the involution $u^* = a^2/u$ always preserves the integer box. It preserves the relaxed M gate and swaps the rational tails. For a relaxed E input it preserves the relaxed gate exactly when

$$K(R) \mid p^2 - 1. \quad (16)$$

For M the exact identity

$$4uG_M(u^*) - pG_M(u) = R(4a + p)$$

shows that the complement has the same gcd with R , because $4u$ and p are units modulo R . Thus a proper M source has a proper M complement and the two rational tails are exchanged.

For E, $R \mid G_E^2$ means $4u \equiv -1 \pmod{K(R)}$, and $4u^* + 1 \equiv 1 - p^2$ there. This proves both directions of (16).

A complement flag followed by an allowed factor k maps a to $(p + R/k)/4$ and decreases it exactly when $k > 1$. This strictness is automatic for a proper M complement, but not for an E complement. For example, $(p, a, R, u, c) = (37, 16, 27, 2, E)$ is proper, whereas its complement $u^* = 128$ has $G_E(u^*) = 513 = 19 \cdot 27$, hence $d^* = 1$; its modulus is $4K(128) = 64$, and its only allowed factor is $k = 1$. The original input is recovered from the flag and old a by applying the same involution. A fixed E word $u = a$ is not counted as two new source words.

6 Composite examples where the square choice and multiplicities matter

At the hard prime $p = 1009$, take $a = 321 = 3 \cdot 107$, $R = 275$, $u = 9$, channel M. The rational tails are $215926/5, 6054/5$ and have sum 44396. Exhaustion of the nine original divisor words $u \mid 321^2$ shows that both full channels of that old shell are empty. Here $d = 5$, $m = 12$ and $k = 25$ gives

$$(a', R', u, h', r', s', \lambda') = (255, 11, 9, 1, 3, 85, 8),$$

$$\boxed{\frac{4}{1009} = \frac{1}{255} + \frac{1}{686120} + \frac{1}{24216}}. \quad (17)$$

This is a genuine composite distinguished denominator, not the preceding prime-shell case.

At $p = 1108801$, take

$$a = 279295 = 5 \cdot 83 \cdot 673, \quad R = 8379 = 19 \cdot 3^2 \cdot 7^2, \quad u = 5, \quad c = M.$$

Its denominator is $d = 3$, its budget modulus is 20 and its entire removal set is

$$\boxed{\mathcal{K} = \{21, 441\}}. \quad (18)$$

The prime 7 is not needed to remove the missing denominator 3, but its residue is needed to preserve the original divisor word. The minimal square factor $q_0 = 3$ fails, whereas $q = 3 \cdot 7$ succeeds. The non-square deletion 21 also succeeds. The two target records are

k	a'	R'	h'	r'	s'	λ'
21	277300	399	5	1	55460	139
441	277205	19	5	1	55441	2918.

The integer coefficient in (8) is two. Its labelled representatives are $v = 7$ and $v = 147$, giving $k = 21$ and $k = 441$. Thus $e_7 + e_{147}$ survives in the labelled source while its group-ring image $2[7]$ vanishes after reduction modulo two. There is one retained original word $u = 5$ and two distinct labelled deletion witnesses and target states.

At $p = 51361$, take $a = 14845$, $R = 8019 = 11 \cdot 3^6$, $u = 5$, channel M. Here $d = 27$, $q_0 = 9$, $t = 27$ and $m = 20$. The only removal is $k = 81$, giving

$$(a', R', h', r', s', \lambda') = (12865, 99, 5, 1, 2573, 26),$$

$$\boxed{\frac{4}{51361} = \frac{1}{12865} + \frac{1}{17179740890} + \frac{1}{6676930}}.$$

The returned residual still contains 3^2 . Removing every square instead would give $R' = 11$, $a' = 12843$, whose square is not divisible by 5. Thus maximal square stripping is not a valid substitute for (10).

The old composite negative control in [3, 4] can also be repaired by a different, explicit map. At $p = 1009, a = 286, R = 135$, its original E word is $u = 7436$, with rational tails

6413/3, 168238642/3. No same-word removal is available. But $K(R) = 45$ divides $p^2 - 1$, so its complement $u^* = 11$ is another relaxed E source. At that source the complete removal set is $\{45\}$, not a square deletion. It gives

$$(a', R', u, h', r', s', \kappa') = (253, 3, 11, 11, 1, 23, 344),$$

$$\boxed{\frac{4}{1009} = \frac{1}{253} + \frac{1}{87032} + \frac{1}{3818056}}.$$

The previously failed map to $R = 15, a = 256$ remains failed. Its replacement changes the original word by the recorded complement and then uses a non-square divisor.

7 The nine-seed list is maximal, even allowing every residual divisor

We first retain the earlier cofactor-capacity estimate.

Lemma 7.1 (Capacity obstruction, with original target retained). *Let $H = 4j - 1 > u > 0$. If $u \nmid j^2$ and a divisor $W \mid j^2$ satisfies $W \equiv u \pmod{H}$, then*

$$H \leq u^2 - 3u + 1. \quad (19)$$

Proof. Write $W = u + nH$ with $n \geq 1$, $V = j^2/W$. The exact equality $16uV = 1 + H(H + 2 - 16nV)$ has a positive final factor congruent to one modulo four. Write it as $4\ell + 1$, $\ell \geq 0$. Comparison gives

$$j = \ell + 4nV, \quad u = (4\ell + 1)n + \ell^2/V.$$

If $\ell = 0$, then $j = 4uV$, contrary to $u \nmid j^2$. Otherwise $w = \ell^2/V$ is a positive integer, and $\ell \leq (u - 2)/4$. Consequently

$$H + 1 = 4\ell + \frac{16\ell^2(u - w)}{(4\ell + 1)w} \leq 4\ell + \frac{16\ell^2(u - 1)}{4\ell + 1} \leq u^2 - 3u + 2.$$

The last function is increasing for $\ell \geq 0$; substitution of $(u - 2)/4$ proves the bound. This is the finite-repair argument of [5], reproduced with the present original target u . $\square$

Theorem 7.2 (Exact universal seed classification). *For a fixed positive integer u , the following are equivalent:*

1. $u \mid 36$;
2. every proper E or M trace source with word u , at every prime $p \equiv 1 \pmod{4}$, admits a same-word residual-divisor return.

For every $u \nmid 36$ there are infinitely many hard primes with an available original M word u and an integral rational first trace, but no such return. Its middle complement has no such return either. At these examples a specified complete original middle cofactor box is empty, although the prime has explicit integer endpoint solutions in both channels.

Proof. The positive implication is Theorem 4.2. For necessity, put $m = 4K(u)$. By (13) there is a unit $b \pmod{m}$ with $b^2 \neq 1$. Replace it by its negative if needed to have $b \equiv 1 \pmod{4}$. Dirichlet's theorem [2, Theorem 18.1] supplies a prime

$$\delta \equiv -b^2 \pmod{m}, \quad \delta > \max(u, u^2 - 3u + 1, 7),$$

and a distinct prime

$$t \equiv b^{-1} \pmod{m}, \quad t > \max(u, 7).$$

Then $\delta \equiv 3 \pmod{4}$, $t \equiv 1 \pmod{4}$, and both primes are coprime to $L = \text{lcm}(840, m)$. Put $R = \delta t^2$ and choose a prime $e \equiv 3 \pmod{4}$ coprime to LR . The three congruences

$$p \equiv 1 \pmod{L}, \quad p \equiv \delta t - 4u \pmod{R}, \quad p \equiv -1 \pmod{e} \quad (20)$$

form one reduced class modulo LR . The middle residue is a unit at both δ and t , since u is coprime to them. Another application of Dirichlet therefore supplies infinitely many primes in this class with $p > \max(R, e, B(u)R)$.

Let $a = (p + R)/4$. Since $R \equiv -b^2 b^{-2} = -1 \pmod{m}$, the integer a is a multiple of $K(u)$, proving the original budget $u \mid a^2$. Also

$$\gcd(R, p + 4u) = \delta t, \quad d = t > 1.$$

This is a proper rational trace source at a hard prime, with no full M gate. The only eligible residual divisors congruent to three modulo four are $\delta, \delta t, R$. The last fails its gate. For the first two, respectively,

$$p + \delta \equiv 1 - b^2 \not\equiv 0 \pmod{m}, \quad p + \delta t \equiv 1 - b \not\equiv 0 \pmod{m}.$$

Their original word budgets fail, so no residual-divisor return exists.

For the middle complement $u^* = a^2/u$, direct valuation comparison gives

$$K(u^*) = a/B(u), \quad 4K(u^*) = (p + R)/B(u) > R.$$

The complement remains a proper trace source with the same exact denominator. Indeed, modulo R ,

$$4u(p + 4u^*) = 4up + 16a^2 \equiv p(p + 4u),$$

and both $4u$ and p are units. Hence $\gcd(R, p + 4u^*) = \gcd(R, p + 4u) = \delta t$ and $d^* = t > 1$. No divisor $1 < k \leq R$ can be one modulo $4K(u^*)$, while $k = 1$ fails $d^* \mid k$. Thus choosing the other original rational-tail orientation does not escape this local obstruction.

The old residual's squarefree part is the prime δ . Put $j = (\delta + 1)/4$. The congruence $\delta + 1 \not\equiv 0 \pmod{4K(u)}$ gives $u \nmid j^2$. A middle cofactor- δ word would require $W \mid j^2$ and $W \equiv -p/4 \equiv u \pmod{\delta}$. The strict size choice of δ contradicts Lemma 7.1; thus the *entire* cofactor box is empty. This is not a claim that the original residual- δ shell, which is a different source, is empty.

Finally $e \mid p + 1$ yields genuine endpoint states. With $c = (p + 1)/e$ and $j_e = (e + 1)/4$, use

$$E : a_E = u_E = (p + e)/4, \quad (h_E, r_E, s_E, \kappa_E) = (a_E, 1, 1, c),$$

$$M : (h_M, r_M, s_M, \lambda_M) = (j_e, 1, c, 1), \quad a_M = j_e c, \quad R_M = c + 1, \quad u_M = j_e.$$

$p \equiv 1 \pmod{4}$ and $e \equiv 3 \pmod{4}$ give $c \equiv 2 \pmod{4}$, so $R_M \equiv 3 \pmod{4}$; also $R_E = e \equiv 3 \pmod{4}$. The inequalities $p > e$, $c \geq 2$, $e \geq 3$ give the original ranges. Their channel-labelled identities follow by multiplication. In particular the M tuple $(j_e c, p j_e c, p j_e)$ is not asserted to increase; its increasing permutation is $(j_e c, p j_e, p j_e c)$. These are counterexamples to the proposed descent predicate, not to ES. $\square$

7.1 A complete hard-prime terminal example

For $u = 8$ choose

$$m = 16, \quad b = 5, \quad \delta = 71, \quad t = 13, \quad e = 11, \quad R = 11999, \quad L = 1680.$$

The reduced prime progression in (20) is

$$p = 48116881 + 221741520n.$$

Its first term is prime, proved by trial division through 6936. It has

$$a = 12032220, \quad d = 13, \quad u = 8,$$

with rational tails

$$\frac{943403747137533690}{13}, \quad \frac{627251660716}{13}.$$

Their sum is 72569567260707262. At the proposed residual 71 the new distinguished denominator is 12029238, whose square is 4 mod 8. The original exponent-three requirement at two is lost by exactly one occurrence. The complement budget modulus is $24064440 > 11999$ and has no proper deletion.

The complete cofactor-71 box is $\text{Div}(18^2)$; its residues are

W	1	2	3	4	6	9	12	18
$W \bmod 71$	1	2	3	4	6	9	12	18
W	27	36	54	81	108	162	324	
$W \bmod 71$	27	36	54	10	37	20	40.	

There is no target residue eight. The separate endpoint middle state has $j_e = 3, c = 4374262$ and denominators $(13122786, p \cdot 13122786, 3p)$.

8 What remains and what the finite execution establishes

The composite-source return is exact; the nine automatic words are a maximal fixed-seed guarantee for this entire residual-divisor method. Beyond them, the complete coefficient test (8), its square subfamily, and the trace complement are useful but not universally occupied. The infinite obstruction retains an actual original word budget and both middle orientations. Enlarging the search over the same residual divisors cannot repair those sources. A new universal arrow must change the word beyond that complement, supply a non-divisor residual through a proved arithmetic construction, or force an original coefficient in the complete source. None is assumed here.

The main verifier enumerates every original square-divisor word at all primes $p \equiv 1 \pmod{4}$ through 3000. It checks all proper trace sources, every return and inverse fibre, actual multiplicities, the unit-square classification through $u = 200$, and the explicit reduced progressions for the 91 nonautomatic seeds through 100. The latter progression records are not 91 asserted prime values. The separate implementation imports neither the main program nor predecessor code: it generates primitive (h, r, s) sources and tests every target residual divisor directly using rational denominators. It independently reproduces the complete original source lists and all output records. The command receipts state the observed outcomes; no independent mathematical review or Lean build is claimed. No finite scan is used as an all-prime occupancy proof.

References

- [1] Elsholtz, C., & Tao, T. (2013). Counting the number of solutions to the Erdős–Straus equation on unit fractions. *Journal of the Australian Mathematical Society*, 94, 50–105. arXiv:1107.1010, §2, Proposition 2.2 and Proposition 2.6.
- [2] Sutherland, A. V. (2017). *18.785 Lecture 18: Dirichlet L-functions, primes in arithmetic progressions*. MIT. Theorem 18.1.
- [3] The Clankers. (2026, September 21). *Prime-shell trace descent and an exact global denominator cutoff*. Preceding programme tranche, sections “Exact rational source”, “Complete prime-shell classification”, and “Actual same-prime middle return”.
- [4] The Clankers. (2026, September 20). *Global trace rigidity and original-source recovery*. TR17–19 and proof.
- [5] The Clankers. (2026, September 20). *Fixed-target cofactor capacity completion*. Theorem “Sharp finite-repair bound”, equations (11)–(12), and proof.